

Network Analysis – Key Terms

1. Node – A point where two or more circuit elements connect.
2. Branch – A path connecting two nodes containing circuit elements.
3. Loop – A closed path in a circuit.
4. Mesh – A loop with no other loops inside it.
5. Circuit – Interconnection of electrical components forming a complete path.
6. Current (I) – Flow of electric charge in a circuit.
7. Voltage (V) – Potential difference between two points.
8. Resistance (R) – Opposition to current flow.
9. Ohm's Law – Relation: $V = IR$.
10. Power (P) – Rate of energy transfer ($P = VI$).
11. Kirchhoff's Current Law (KCL) – Sum of currents at a node is zero.
12. Kirchhoff's Voltage Law (KVL) – Sum of voltages in a loop is zero.
13. Impedance (Z) – Total opposition in AC circuits.
14. Admittance (Y) – Reciprocal of impedance.
15. Capacitance (C) – Ability to store electric charge.
16. Inductance (L) – Ability to store energy in a magnetic field.
17. Reactance (X) – Opposition due to capacitance or inductance.
18. Frequency (f) – Number of cycles per second.
19. Phase – Relative position of waveforms.
20. AC (Alternating Current) – Current that changes direction periodically.
21. DC (Direct Current) – Current flowing in one direction.
22. Thevenin's Theorem – Reduces circuit to voltage source and resistance.
23. Norton's Theorem – Reduces circuit to current source and resistance.
24. Superposition Theorem – Analyzes effect of each source separately.
25. Maximum Power Transfer – Power is max when load equals source resistance.
26. Time Constant (τ) – Time for system to reach ~63% response.
27. Transient Response – Temporary behavior before steady state.
28. Steady State – Condition after transients settle.
29. Network Function – Output-input relationship in frequency domain.
30. Transfer Function – Ratio of output to input in Laplace domain.